

Math 162

Problem set 5 solutions

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1. Using the first fundamental theorem of calculus, the hypothesis $f(x) = \int_0^x f(t) dt$ implies that $f' = f$. As shown in class (theorem characterizing the exponential function), it follows that $f(x) = C \exp(x)$ for some $C \in \mathbb{R}$. Apply the hypothesis on f again to obtain

$$C \exp(x) = \int_0^x C \exp(t) dt = C \exp(t) \Big|_{t=0}^{t=x} = C(\exp(x) - 1).$$

Substituting $x = 0$, we see that $C = 0$ and hence $f = 0$.

2. (a) My hint covers the case $n = 0$. If $n \geq 1$ then L'Hôpital's rule applies, since $\lim_{x \rightarrow \infty} \log x \rightarrow \infty$ by the case $n = 0$, and we proved last quarter that $\lim_{x \rightarrow \infty} x^n = \infty$. We therefore have

$$\lim_{x \rightarrow \infty} \frac{\log x}{x^n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{x}}{nx^{n-1}} = \lim_{n \rightarrow \infty} \frac{1}{nx^n} = 0.$$

- (b) By definition $x^x = \exp(x \log x)$ for $x > 0$. Observe that

$$\lim_{x \rightarrow 0^+} x \log x = \lim_{x \rightarrow 0^+} \frac{\log x}{1/x} = \lim_{x \rightarrow \infty} \frac{\log(1/x)}{x} = - \lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$$

by the case $n = 1$ of part (a). Since $\exp$ is continuous everywhere and in particular at 0, it follows that

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} \exp(x \log x) = \exp 0 = 1.$$

- (c) By definition we have $(1 + \frac{1}{x})^x = \exp(x \log(1 + \frac{1}{x}))$. Note that

$$\lim_{x \rightarrow \infty} x \log(1 + \frac{1}{x}) = \lim_{x \rightarrow 0^+} \frac{\log(1 + x)}{x}.$$

Since $\log$ is continuous we have $\lim_{x \rightarrow 0} \log(1 + x) = \log 1 = 0$, so L'Hôpital's rule applies and we obtain

$$\lim_{x \rightarrow 0^+} \frac{\log(1 + x)}{x} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+x}}{1} = 1.$$

Finally, we deduce that

$$\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x = \exp(1) = e$$

using the fact that $\exp$ is continuous at 1.

3. (a) Let $u := \arctan x$. Then we have $x = \tan u$ and hence $x \cos u = \sin u$. Squaring this identity and using the Pythagorean theorem, we obtain

$$\sin^2 u = x^2 \cos^2 u = x^2(1 - \sin^2 u).$$

Rearranging yields

$$\sin^2 u = \frac{x^2}{1 + x^2}.$$

Take the square root of both sides to obtain

$$|\sin(\arctan x)| = |\sin u| = \frac{|x|}{\sqrt{1+x^2}}.$$

Note that if $x \geq 0$ then $0 \leq \arctan x < \frac{\pi}{2}$, and hence $\sin(\arctan x) \geq 0$. Similarly, if $x < 0$ then $-\frac{\pi}{2} < \arctan x < 0$, which implies that $\sin(\arctan x) < 0$. It follows that

$$\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}.$$

As for $\cos(\arctan x)$, similarly to the above the identity $x \cos u = \sin u$ yields

$$|\cos(\arctan x)| = |\cos u| = \frac{1}{\sqrt{1+x^2}}.$$

Since $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$ for all $x \in \mathbb{R}$, we have $\cos(\arctan x) > 0$ and hence

$$\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}.$$

- (b) Recall that $\tan$ is strictly increasing and hence injective on each interval $((2k-1)\frac{\pi}{2}, (2k+1)\frac{\pi}{2})$ for $k \in \mathbb{Z}$. Moreover, we have $\tan(x+\pi) = \tan x$ for any $x \in \mathbb{R}$. Thus $x = \tan \frac{u}{2}$ implies that $\frac{u}{2} = k\pi + \arctan x$ for some $k \in \mathbb{Z}$. We obtain $u = 2k\pi + 2\arctan x$ and hence

$$\sin u = \sin(2k\pi + 2\arctan x) = \sin(2\arctan x) = 2\sin(\arctan x)\cos(\arctan x)$$

by the double-angle identity (a special case of the angle addition formula). Now apply part (a), which yields

$$2\sin(\arctan x)\cos(\arctan x) = 2 \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} = \frac{2x}{1+x^2}.$$

Similarly, we have

$$\cos u = \cos(2k\pi + 2\arctan x) = \cos(2\arctan x) = \cos^2(\arctan x) - \sin^2(\arctan x)$$

by the double-angle identity, and then

$$\cos^2(\arctan x) - \sin^2(\arctan x) = \frac{1}{1+x^2} - \frac{x^2}{1+x^2} = \frac{1-x^2}{1+x^2}$$

by part (a).

4. We claim that $\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$, which we will prove using the limit $\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \infty$ from class. Fix $\epsilon > 0$, and put $M := \tan(\frac{\pi}{2} - \epsilon)$. If $x > M$, then since $\arctan$ is increasing, we have $\arctan x > \frac{\pi}{2} - \epsilon$. Since $\arctan x < \frac{\pi}{2}$ for any $x \in \mathbb{R}$, it follows that

$$|\arctan x - \frac{\pi}{2}| = \frac{\pi}{2} - \arctan x < \epsilon$$

as needed.

We saw in class that $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$, so

$$\int_0^\infty \frac{dx}{1+x^2} = \lim_{N \rightarrow \infty} \int_0^N \frac{dx}{1+x^2} = \lim_{N \rightarrow \infty} \arctan N = \frac{\pi}{2}.$$

On the other hand, we have

$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{N \rightarrow \infty} \int_{-N}^0 \frac{dx}{1+x^2} = \lim_{N \rightarrow \infty} \int_0^N \frac{dx}{1+x^2} = \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$$

because $\frac{1}{1+x^2}$ is even. Thus

$$\int_{-\infty}^\infty \frac{dx}{1+x^2} = \int_0^\infty \frac{dx}{1+x^2} + \int_{-\infty}^0 \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

5. First, we will prove using induction that for any $k \geq 0$, there exist polynomials p and q such that

$$f^{(k)}(x) = e^{-1/x^2} \left(p\left(\frac{1}{x}\right) \cos \frac{1}{x} + q\left(\frac{1}{x}\right) \sin \frac{1}{x} \right)$$

for all $x \neq 0$. Of course when $k = 0$, we can take $p(x) = 0$ and $q(x) = 1$. For the inductive step, suppose that for some k we have proved that

$$f^{(k)}(x) = e^{-1/x^2} \left(p\left(\frac{1}{x}\right) \cos \frac{1}{x} + q\left(\frac{1}{x}\right) \sin \frac{1}{x} \right)$$

for all $x \neq 0$. Then we have

$$\begin{aligned} f^{(k+1)}(x) &= \frac{2}{x^3} e^{-1/x^2} \left(p\left(\frac{1}{x}\right) \cos \frac{1}{x} + q\left(\frac{1}{x}\right) \sin \frac{1}{x} \right) + e^{-1/x^2} \left(-\frac{1}{x^2} p'\left(\frac{1}{x}\right) \cos \frac{1}{x} \right. \\ &\quad \left. + p\left(\frac{1}{x}\right) \cdot \frac{1}{x^2} \sin \frac{1}{x} - \frac{1}{x^2} q'\left(\frac{1}{x}\right) \sin \frac{1}{x} - q\left(\frac{1}{x}\right) \cdot \frac{1}{x^2} \cos \frac{1}{x} \right) \\ &= e^{-1/x^2} \left(\left(\frac{2}{x^3} p\left(\frac{1}{x}\right) - \frac{1}{x^2} p'\left(\frac{1}{x}\right) - q\left(\frac{1}{x}\right) \right) \cos \frac{1}{x} + \left(\frac{2}{x^3} q\left(\frac{1}{x}\right) + \frac{1}{x^2} p\left(\frac{1}{x}\right) - \frac{1}{x^2} q'\left(\frac{1}{x}\right) \right) \sin \frac{1}{x} \right), \end{aligned}$$

and the coefficients of $\cos \frac{1}{x}$ and $\sin \frac{1}{x}$ are polynomials in $\frac{1}{x}$ as desired.

For the next step, we will need to prove that $\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^n} = 0$ for any $n \in \mathbb{N}$. Since

$$\left| \frac{e^{-1/x^2}}{x^n} \right| = \frac{e^{-1/x^2}}{|x|^n}$$

and

$$\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x^n} = \lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{|x|^n},$$

by the squeeze theorem it suffices to show that $\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x^n} = 0$. For this, observe that

$$\lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x^n} = \lim_{x \rightarrow \infty} \frac{e^{-x^2}}{x^{-n}} = \lim_{x \rightarrow \infty} \frac{x^n}{e^{x^2}}.$$

We have $x^2 \geq x$ for all $x \geq 1$ and hence

$$\frac{x^n}{e^{x^2}} \leq \frac{x^n}{e^x}.$$

We saw in class that $\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$, which implies that $\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$ and hence $\lim_{x \rightarrow \infty} \frac{x^n}{e^{x^2}} = 0$ by the squeeze theorem.

Finally, we prove that $f^{(k)}(0) = 0$ for all $k \geq 0$ by induction. The case $k = 0$ just says that $f(0) = 0$, which is part of the definition of f . Assuming that $f^{(k)}(0) = 0$ for the inductive step, we have

$$f^{(k+1)}(0) = \lim_{h \rightarrow 0} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} e^{-1/h^2} \left(p\left(\frac{1}{h}\right) \cos \frac{1}{h} + q\left(\frac{1}{h}\right) \sin \frac{1}{h} \right)$$

using the inductive hypothesis and the previously established formula for $f^{(k)}(h)$, $h \neq 0$. Note that

$$\left| \frac{1}{h} e^{-1/h^2} \left(p\left(\frac{1}{h}\right) \cos \frac{1}{h} + q\left(\frac{1}{h}\right) \sin \frac{1}{h} \right) \right| \leq \frac{1}{|h|} e^{-1/h^2} (|p(\frac{1}{h})| + |q(\frac{1}{h})|).$$

Now $p(\frac{1}{h})$ and $q(\frac{1}{h})$ are sums of terms of the form $\frac{c}{h^n}$ where $c \in \mathbb{R}$, from which it follows that

$$\frac{1}{|h|} e^{-1/h^2} (|p(\frac{1}{h})| + |q(\frac{1}{h})|) \leq \sum_{n=1}^d e^{-1/h^2} \cdot \frac{c_n}{|h|^n}$$

for some $c_n \geq 0$ and $d \in \mathbb{N}$. We have

$$\lim_{h \rightarrow 0} \frac{e^{-1/h^2}}{|h|^n} = \lim_{h \rightarrow 0^+} \frac{e^{-1/h^2}}{h^n} = 0$$

by the limit established above, so each term of the above sum approaches 0 as $h \rightarrow 0$. We conclude that $f^{(k+1)}(0) = 0$ by the squeeze theorem, which completes the induction.